POLYDIM CONSTITUTION

Formal Axiomatics and Topological Foundations for Cognitive Programming

POPL 2027 DRAFT

The Epistemological Shift

From Algorithms to Processes

The classical paradigm is predicated on executing deterministic algorithms within discrete machines. **Cognitive Programming** shifts the focus to the constrained evolution of internal representational manifolds.

Representational Geometry

Execution is no longer a path of instruction sequences, but a directed trajectory designed to navigate topological latent vectors toward a precise semantic objective.

Classical Paradigm



Cognitive Paradigm



The Cognitive Program Tuple

Mathematical Encapsulation

A Cognitive Program is formally defined as an axiomatic system governed by a 6-tuple representational framework.

$$P = (S, G, T, O, C, \Pi)$$

The mathematical architecture guarantees execution safety and logical consistency by verifying constraints continuously on the state manifold.

- **State (S):** The complete latent representational manifold.
- **Goal (G):** The semantic target delineating valid trajectories.
- **Transformation (T):** Space endomorphism $T : S \to S$.
- **Observer (O):** Mapping functor from state space to semantics.
- **Constraint (C):** Predicate invariant $C(S) \equiv \text{True}$.
- **Policy (Π):** Meta-rules governing selection criteria for T.

Axiomatic Theory of Polydimensions

Polydimension Definition

The core abstraction coordinates high-entropy transformations with absolute topological preservation across dynamic states.

$$\mathbb{P} = (D , \Pi_P , T , \Omega)$$

D: Continuous or discrete topological dimensions.

Π_P: Projection functors preserving homotopic invariance.

T: Admissible transformations (∞-groupoid of identity).

Ω: Topos subobject classifier defining intuitionistic logic.



The Seven Axioms of Execution

Existence & Obs

Axiom I & II: Every cognitive program possesses a non-empty representational state S and admits an observation functor O.

Closure & Homotopy

Axiom III & IV: Transformations stay within boundary D. The ∞-groupoid is composable and weakly invertible up to higher homotopy.

Teleology & Utility

Axiom V, VI, VII: Execution is teleological (goal-bound), continuously checked for constraints, with a dynamic policy Π.

Topos & Category Theory Foundations



Functorial Correctness

POLYDIM guarantees identity validation using relative coordinates without referencing absolute anchors.

The Yoneda Lemma

$$\mathrm{Nat}(\mathrm{Hom}(-, X), F) \cong F(X)$$

Objects are entirely determined by their morphisms. Grothendieck Topos Theory ensures invalid computational states gracefully collapse to the empty fiber ($\varnothing$) without losing topological continuity.

Geometric Algebra

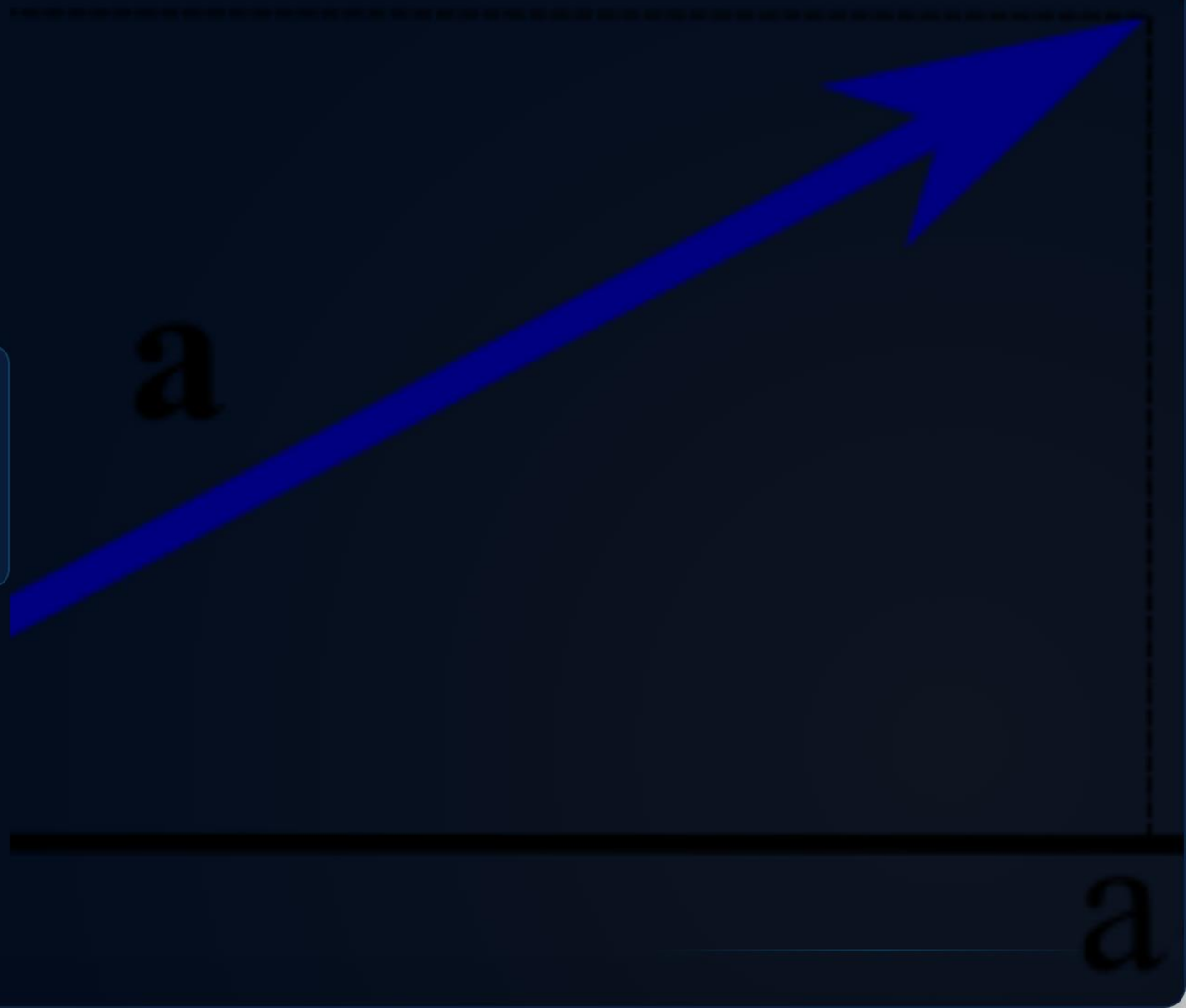
Subspace Invariance

We leverage multi-dimensional Geometric Algebras to represent rotations and projections within cognitive execution loops.

Hodge Decomposition

$$\omega = d\alpha + \delta\beta + \gamma$$

Cartan-Dieudonné Theorem: Bounds orthogonal complexity. **Hodge Decomposition:** Separates divergence-free elements from rotational noise, purging hallucinations.



Topology & Information Geometry

Freudenthal Suspension

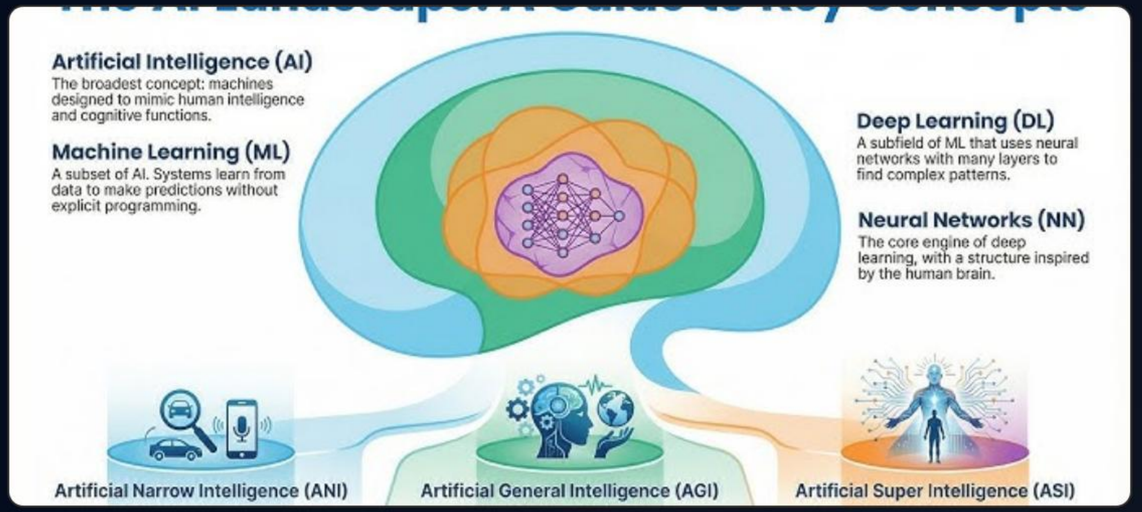
Enables the lossless elevation of low-dimensional topological invariants into highly dimensional multivector spaces, creating robust nominal categories from continuous patterns.

Fisher-Rao Information Metric

Provides the Riemannian structure for the manifold of probability distributions. Optimizes update trajectories by ensuring they traverse natural geodesics rather than arbitrary Euclidean steps, maintaining extreme stability.

Geometric Attractors

Information Geometry defines the "semantic gravity" pulling representations toward stable, predictable attractors, avoiding objective drift.



Derived Structural Theorems

Theorem 1

Covariant Morphism: Every admissible transformation $T \in \mathbb{T}$ induces a strictly covariant morphism between compatible polydimensions, guaranteeing logical alignment across different substrates.

Theorem 2

GeoID Preservation: The structural identity of representational paths remains homotopically invariant under any valid computational projection.

$$\forall T \in \mathbb{T}, \text{GeoID}(T(x)) \cong \text{GeoID}(x)$$

Projection & Tomography Theorems

Theorem 3: Projection Commutativity

Under strict semantic isotropy, sequential projection pathways commute. In real-world environments, this commutativity is elegantly bounded by the torsion tensor of the latent representation manifold.

$$\text{PROJECT}_B\ (\ \text{PROJECT}_A\ (\ x\ )\ )\ \simeq\ \text{PROJECT}_A\ (\ \text{PROJECT}_B\ (\ x\ )\ )$$

Theorem 4: Semantic Tomography

Given sufficient orthogonal projection coordinates, the underlying high-dimensional continuous tensor can be reconstructed with near-zero loss. This is mathematically isomorphic to the Radon Transform in integral geometry.

Theorem of Cognitive Preservation

Zero

Objective Drift



Mathematical Goal Preservation

The primary challenge of Cognitive Programming is proving that an execution preserves the target objectives G across varying substrates.

Compilation is not syntax translation; it is the **homomorphic preservation** of cognitive pathways. Resolving this theorem equates to the complete eradication of semantic hallucinations.

> *Conjecture: The translation from geometric semantics to discrete nominal types perfectly preserves all topological limits of the underlying topos.*

Polydim Constitution

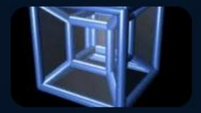
Bridging Topological Idealism with Demonstrable Computational Rigor.

> "The homomorphic translation of thought requires structural preservation, not merely physical equivalence."

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Image Sources



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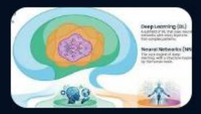
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